$$\begin{aligned}\cos 3x &= \cos(2x + x) \\ &= \cos 2x \cos x - \sin 2x \sin x \\ &= (2\cos^2 x - 1)\cos x - 2\sin^2 x \cos x \\ &= (2\cos^2 x - 1)\cos x - 2(1 - \cos^2 x)\cos x\end{aligned}$$

$$\boxed{\cos 3x = 4\cos^3 x - 3\cos x}$$

$$\begin{aligned}\sin 3x &= \sin(2x + x) \\ &= \sin 2x \cos x + \cos 2x \sin x \\ &= 2\sin x \cos^2 x + (1 - 2\sin^2 x)\sin x \\ &= 2\sin x(1 - \sin^2 x) + (1 - 2\sin^2 x)\sin x\end{aligned}$$

$$\boxed{\sin 3x = 3\sin x - 4\sin^3 x}$$

(i)

$$7\sin x - 8\sin^3 x = \sin x + 2(3\sin x - 4\sin^3 x)$$

$$= \sin x + 2\sin 3x$$

$$I(\alpha) = \int_0^{\alpha} (\sin x + 2\sin 3x)\, dx$$

$$= \left[-\cos x - \frac{2}{3}\cos 3x\right]_0^{\alpha}$$

$$= -\cos\alpha - \frac{2}{3}\cos 3\alpha + \frac{5}{3}$$

Let $c = \cos\alpha$

$$\cos 3\alpha = 4c^3 - 3c$$

$$I(\alpha) = -c - \frac{2}{3}(4c^3 - 3c) + \frac{5}{3}$$

$$\boxed{I(\alpha) = -\frac{8}{3}c^3 + c + \frac{5}{3}}$$

For $c = 1$

$$I(\alpha) = -\frac{8}{3} + 1 + \frac{5}{3} = 0$$

$$\boxed{c = 1}$$

(ii)

Eustace obtains

$$I_E(\alpha) = \left[\frac{7}{2}\sin^2 x - 2\sin^4 x\right]_0^{\alpha}$$

$$= \frac{7}{2}\sin^2\alpha - 2\sin^4\alpha$$

$$= \frac{7}{2}(1-c^2) - 2(1-c^2)^2$$

$$= \frac{3}{2} + \frac{1}{2}c^2 - 2c^4$$

For $\cos\beta = -\dfrac{1}{6}$

$$I(\beta) = -\frac{8}{3}\left(-\frac{1}{6}\right)^3 - \frac{1}{6} + \frac{5}{3}$$

$$= \frac{245}{162}$$

$$I_E(\beta) = \frac{3}{2} + \frac{1}{2}\left(-\frac{1}{6}\right)^2 - 2\left(-\frac{1}{6}\right)^4$$

$$= \frac{245}{162}$$

$$\boxed{I_E(\beta) = I(\beta)}$$

For the correct value of $I(\alpha)$

$$\frac{3}{2} + \frac{1}{2}c^2 - 2c^4 = -\frac{8}{3}c^3 + c + \frac{5}{3}$$

$$12c^4 - 16c^3 - 3c^2 + 6c + 1 = 0$$

$$(c-1)^2(2c+1)(6c+1) = 0$$

$$c = 1, \qquad c = -\frac{1}{2}, \qquad c = -\frac{1}{6}$$

Since $c = \cos\alpha$

$$\boxed{\alpha = 2k\pi, \qquad \alpha = 2k\pi \pm \frac{2\pi}{3}, \qquad \alpha = 2k\pi \pm \arccos\left(-\frac{1}{6}\right), \quad k \in \mathbb{Z}}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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